

Fake News Detector

User Guide

Installation, usage, troubleshooting, and responsible-use guidance for the Streamlit web application

Fake News Detection Using Advanced NLP – Internship Project

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Table of Contents

1. Overview

The Fake News Detector is a Streamlit web application that screens news article text for linguistic patterns associated with fabricated or unreliable reporting. You paste in an article's title and body, and it returns a REAL or FAKE prediction along with a confidence score, using a classical machine learning model trained on the ISOT Fake and Real News dataset.

The app is one piece of a larger project: the model behind it was trained and evaluated through a full pipeline of exploratory data analysis, text preprocessing, TF-IDF/Word2Vec feature engineering, and a six-way comparison across Naive Bayes, SVM, and Random Forest. This guide covers only the app itself – installing it, running it, reading its output correctly, and fixing the most common setup problems.

Important: This tool detects writing-style patterns, not factual accuracy. It does not check any claim in an article against real-world facts or an external database. Treat every prediction as a screening signal to guide further review, never as a final verdict.

2. System Requirements

Requirement	Details
Python	3.10 or later
Key packages	streamlit, scikit-learn, nltk, gensim (only if the saved model uses Word2Vec), numpy
Trained model files	best_model.pkl, best_model_metadata.json, and either tfidf_vectorizer.pkl or word2vec.model – all inside a models/ folder
Browser	Any modern browser (Chrome, Edge, Firefox) – the app opens automatically at http://localhost:8501
Disk layout	webapp/ (containing app.py) and models/ must be sibling folders under the same project root

3. Installation and Setup

3.1 Folder Structure

Before launching the app for the first time, confirm your project folder looks like this (folder names and nesting matter – see Section 7 for what goes wrong if they don't match):

```
Fake-News-Detection/  
  models/  
    best_model.pkl
```

```

best_model_metadata.json
tfidf_vectorizer.pkl    (or word2vec.model)
all_trained_models.pkl
webapp/
  app.py
notebooks/
dataset/
requirements.txt

```

3.2 Install Dependencies

From the project root, create and activate a virtual environment, then install the required packages:

```

python -m venv venv
source venv/bin/activate    # Windows: venv\Scripts\activate
pip install -r requirements.txt

```

3.3 Confirm the Model Files Exist

The app cannot run without a trained model. If `models/best_model.pkl` doesn't exist yet, run notebooks 06 (Model Training), 07 (Model Evaluation), and 08 (Save Best Model) first — 08 is the one that actually writes `best_model.pkl`, `best_model_metadata.json`, and copies the matching vectorizer into `models/`.

4. Launching the App

Open a terminal, activate your virtual environment, and run one of the following. Both work correctly regardless of which folder you launch from, since the app resolves its own file paths based on its own location on disk rather than the folder you happened to run the command from:

```

# From inside webapp/
cd webapp
streamlit run app.py
# --- OR, equivalently, from the project root ---
streamlit run webapp/app.py

```

A browser tab opens automatically, usually at `http://localhost:8501`. If it doesn't open on its own, copy the local URL printed in the terminal into your browser manually.

5. Using the App: Step by Step

1. Wait for the sidebar to finish loading. The top-right corner shows “CONNECTING” while Streamlit starts up. Once the sidebar shows a model name and a test F1-score,

the app is ready to use.

2. Paste the full article text — title and body — into the large text box in the center of the page. Longer, more complete text gives the model more signal to work with than a single sentence or headline alone.
3. Click “Analyze Article.” The app runs your pasted text through the exact same cleaning pipeline used during training (removing formatting noise and boilerplate, tokenizing, removing stopwords, lemmatizing) before scoring it — you don’t need to clean or format the text yourself.
4. Read the result card. A green card with a checkmark means the model predicts REAL; an orange card with a warning icon means FAKE.
5. Read the confidence percentage underneath the label. This is the model’s confidence in that specific prediction, not a general accuracy figure — see Section 6 for how to interpret it.
6. Open the “How was this determined?” expander below the result for a plain-language explanation of which model produced the prediction and how many meaningful words survived the cleaning step.
7. If you see a warning instead of a result (text too short, or no usable words after cleaning), follow the on-screen guidance — usually pasting a longer excerpt resolves it immediately.

6. Understanding the Results

6.1 Label and Confidence

The confidence score matters as much as the label itself. A REAL prediction at 52% confidence is a near-coin-flip; the same label at 97% confidence reflects a much stronger pattern match. As a rough guide:

Confidence range	How to treat it
90–100%	Strong pattern match — still not proof, but a confident signal
70–90%	Reasonably confident, worth trusting as a first-pass signal
50–70%	Weak signal — treat as inconclusive rather than decisive

6.2 What the Model Is and Isn't Doing

The model compares the writing style of your pasted text — vocabulary, phrasing, structural patterns — against what it learned from thousands of labeled real and fake articles during

training. It does not look up any fact in your text against an external source, a search engine, or a knowledge base. Two articles making the same false claim, one written in a measured, well-sourced style and one written sensationally, can receive different predictions from this model even though both are equally false – because it is scoring style, not truth.

7. Troubleshooting

7.1 “Model files not found”

This is the most common issue, and it's almost always a folder-location problem rather than a missing file. Expand “Debug: file paths” in the sidebar – it shows the exact folder the app is checking and a checkmark or cross next to each required file.

Symptom	Likely cause	Fix
Debug panel shows the wrong folder entirely	app.py is an old copy without the <code>__file__</code> -based path fix	Confirm the file contains the line <code>BASE_DIR = Path(__file__).resolve().parent</code> ; if not, replace it with the current version
Debug panel shows the right folder, but files are missing	Notebooks 06–08 haven't been run yet, or models/ is misspelled/misplaced	Run 08_Save_Best_Model.ipynb, and confirm models/ sits directly next to webapp/, not inside it or elsewhere
Fix applied but error persists	Streamlit is still running the old process, or a stale cached model	Stop the server fully (Ctrl+C, not just a browser refresh), then in the browser menu choose “Clear cache” before restarting

7.2 App Loads but Predictions Seem Off

- Very short text: predictions on fewer than ~15 words are inherently less reliable – this isn't a bug, it's a genuine limitation of having little linguistic signal to work with.
- Unfamiliar topics: the model was trained on a historical news snapshot; entirely new topics, slang, or breaking events it never saw in training may be handled less reliably.
- Mixed-language or non-English text: the model and its preprocessing assume English text.

7.3 App Won't Start At All

- Check the terminal output for a missing-package error and re-run `pip install -r requirements.txt`.
- Confirm you're using Python 3.10+ and that your virtual environment is actually activated (the terminal prompt should show its name, e.g. `(venv)`).

8. Test Articles

Use these two short articles to sanity-check the app right after installation, before trying real articles you're unsure about.

8.1 Expected REAL

Title: City Council Approves Budget for Road Repairs Ahead of Winter Season

The city council voted on Tuesday to approve a \$4.2 million budget allocation for road repairs and pothole maintenance ahead of the upcoming winter season. The funding will be distributed across twelve districts, with priority given to routes used by public transit and emergency services. According to the public works department, crews will begin repaving sections of the downtown corridor next month, weather permitting. Officials said the schedule was developed based on a survey conducted over the summer that identified more than 300 road segments in need of repair. The council also approved a smaller allocation for sidewalk accessibility upgrades, which will be completed in phases over the next two years.

8.2 Expected FAKE

Title: Scientists Confirm Drinking Coffee Makes People Instantly Fluent in Five Languages

Researchers at the so-called "Global Cognitive Enhancement Institute" have announced a revolutionary discovery claiming that drinking two cups of a newly developed coffee blend instantly enables people to speak five foreign languages fluently. According to the report, over 10,000 volunteers demonstrated perfect communication skills within just fifteen minutes of consuming the beverage.

The institute further claimed the coffee contains "quantum memory molecules" that transfer language knowledge directly to the brain without any learning process. No peer-reviewed study, university affiliation, or clinical trial data was provided to support these claims.

Note: Neither result is guaranteed. If the model gets either of these wrong, that's a genuine data point about its behavior worth noting — not a sign the app is broken.

9. Limitations and Responsible Use

- This is a content-based screening signal, not a fact-checking authority — always verify important claims against a trusted, independent source.
- The model was trained on one historical, predominantly U.S., English-language dataset; its reliability on other regions, languages, or time periods is unproven.
- Do not use this tool as the sole basis for a consequential decision (e.g. removing content, accusing a source of dishonesty) — pair it with human review.
- Re-run the training pipeline periodically if this app is used beyond a demo/portfolio context, since news vocabulary and topics drift over time.

10. Frequently Asked Questions

Can I run this without the internet?

Yes, once installed. The app itself needs no internet connection at runtime; it only needs the model files already saved locally in `models/`.

Can multiple people use it at once?

Streamlit serves one app instance per machine by default; for multiple simultaneous users, it would need to be deployed to a shared server rather than run locally – outside the scope of this guide.

How do I update the model later?

Re-run the training and evaluation notebooks with new or updated data, then re-run `08_Save_Best_Model.ipynb`. The app automatically reads whichever model and feature type `best_model_metadata.json` points to – no changes to `app.py` are needed.